ROBERT J. **PARK** · ENVIRONMENT · CARBON · PUBLIC AFFAIRS

The Land Repair Helix

Site plan: Brownfield

Site name: _____ Date: _____ Prepared by: _____

Compacted, tired soil → Feed the soil life (Soil)

Compacted soil with little organic matter holds less water, drains badly, and needs more fertiliser to grow anything.

- ☐ £ Stop heavy machinery and livestock on wet ground
- ☐ £ Add compost, farmyard manure or digestate to build organic matter
- ☐ ££ Sow deep-rooting cover crops or herbal leys (chicory, plantain) to break up compaction
- ☐ ££ Cut down on ploughing; direct drilling where it suits the soil

NI: Soil sampling is often supported through DAERA schemes. Check what's open before paying for testing.

Bare soil over winter → Keep it green (Soil)

Bare soil erodes in heavy rain and carries sediment and phosphorus into rivers and loughs.

- ☐ £ Leave stubbles, or sow a winter cover crop
- ☐ £ Keep grass margins at field edges and gateways
- ☐ ££ Plant hedges or grass strips across slopes to slow runoff

Flooding and fast runoff → Slow the flow (Water)

Hard surfaces and drained land send water downstream fast, causing floods and washing pollution off the land.

- ☐ £ Break up hard surfaces with gravel, grass-crete or permeable paving
- ☐ ££ Dig swales, rain gardens and small ponds to hold water back
- ☐ ££ Install leaky dams of logs or brash in ditches and small streams
- ☐ £££ Add sustainable drainage (SuDS) when sites are redeveloped

NI: Work in or near a watercourse may need consent from DfI Rivers. Check before building dams or culverts.

Metals in the soil (lead, zinc, copper) → Lock them in, or let plants lift them (Contamination)

Old industry, made ground, and lead from paint or fuel can leave metals that harm people, especially children, and wildlife.

- ☐ £ Test before growing food or creating play areas
- ☐ £ Keep a clean cover: a capping layer of clean soil or a membrane
- ☐ ££ Phytostabilisation: willows and grasses hold metals in place and stop dust
- ☐ ££ Phytoextraction: plants such as Indian mustard take up some metals. It's slow and the harvested plants need proper disposal
- ☐ £££ Remove the hotspots by excavation, under a remediation strategy

NI: NI's contaminated land regime (Part III) isn't in force, so land contamination is dealt with through planning, using LCRM.
Take care: Phytoremediation is slow and only partial. Use it as part of a plan agreed with the planning authority and environmental health, not on its own.

Old fuel and oil in the ground → Let microbes digest it (Contamination)

Leaking tanks and old spills leave hydrocarbons that smell, kill plants and can reach groundwater.

- ☐ £ Find and stop the source: old tanks, pipework, drums
- ☐ ££ Biopiles or windrows: dig out, aerate and add nutrients so bacteria break down the oil
- ☐ ££ Plant deep-rooting vegetation to boost microbes around the roots
- ☐ £££ Excavate and dispose, or treat on site, for heavy contamination

NI: Treating contaminated soil on site normally needs a permit or waste exemption from NIEA.

Forever chemicals (PFAS) → Stop the source, then contain (Contamination)

PFAS don't break down naturally. They come from firefighting foam, coatings and some industrial processes, and spread in water.

- ☐ £ Stop using PFAS firefighting foam; switch to fluorine-free foam and dispose of old stock properly
- ☐ ££ Contain hotspots: immobilise PFAS in soil with activated carbon or specialist sorbents mixed in
- ☐ £££ Treat water with granular activated carbon, ion exchange or reverse osmosis
- ☐ £££ Excavate and destroy by high-temperature treatment

NI: EU REACH applies to goods in NI. The EU ban on PFAS in firefighting foam applies from 23 October 2030.
Take care: No plant or cheap fix removes PFAS. Anything sold as 'natural PFAS removal' should be treated with caution. Testing and treatment need specialists.

Diesel, oil or paraffin spilled on land → Contain, soak up, dispose properly (Spills)

A few litres of oil can pollute a large area of soil, a drain or a stream.

- ☐ £ Stop the leak and block nearby drains with covers or sandbags
- ☐ £ Soak it up with sawdust, wood shavings, straw, coir, cork granules or wool
- ☐ £ Clay granules (cat litter) work well on hard surfaces
- ☐ £ Keep a ready-made spill kit beside every fuel store and generator
- ☐ ££ Dig out badly soaked soil, then let microbes treat it in a biopile

NI: If oil reaches a drain, watercourse or groundwater, ring the NIEA Incident Hotline on 0800 80 70 60. Used sorbents are hazardous waste and must go via a registered carrier.
Take care: Don't use peat, which destroys bogs, or washing-up liquid and detergents, which spread the oil into the water.

Brownfield seen as wasteland → Keep the mosaic (Rewilding)

Bare ground, rubble and patchy plants on brownfield sites can support more insects than tidy grassland.

- ☐ £ Survey before 'greening': rare insects and plants may already be there
- ☐ £ Keep bare patches, rubble piles and shallow pools
- ☐ ££ Only spread topsoil and seed where it's needed
- ☐ ££ Control invasive species such as Japanese knotweed with a specialist

NI: Japanese knotweed waste is controlled waste. It's an offence to cause it to grow in the wild.
Take care: Check for contamination before disturbing made ground.

Few pollinators → Flowers from spring to autumn (Rewilding)

Bees and hoverflies need food from March to October, and nesting spots.

- ☐ £ Leave dandelions and clover in amenity grass
- ☐ £ Plant spring-flowering willow, hawthorn and blackthorn, and late-flowering ivy
- ☐ £ Leave bare south-facing banks for mining bees
- ☐ £ Cut pesticide use

NI: The All-Ireland Pollinator Plan has free guides for farms, councils, businesses and schools.

Trees in the wrong place → Right tree, right place (Air and carbon)

Trees planted on species-rich grassland, peat or nesting habitat can do more harm than good.

- ☐ £ Plant native broadleaves along watercourses for shade; cooler water helps reduce algal blooms
- ☐ £ Plant hedgerow trees and shelterbelts rather than blocks on good habitat
- ☐ ££ Let woodland regenerate naturally where there are seed sources

NI: Check for priority habitat and protected sites before planting.

Invasive plants spreading → Catch them early, treat properly (Rewilding)

Japanese knotweed, Himalayan balsam and rhododendron crowd out native plants, damage structures and spread along rivers and roads.

- ☐ £ Learn to recognise them and map where they are
- ☐ £ Pull Himalayan balsam before it sets seed (June to July), and repeat each year
- ☐ ££ Treat knotweed with a specialist herbicide programme over several seasons
- ☐ £££ Excavate knotweed where development needs it, under a management plan

NI: It's an offence in NI to plant certain invasive species, or cause them to grow, in the wild. Knotweed waste is controlled waste and must go to an authorised site.
Take care: Never strim or flail knotweed: tiny fragments regrow.

Litter and fly-tipping → Design it out, report it fast (Rewilding)

Litter and dumped waste blight places, harm wildlife and signal neglect, which draws more dumping.

- ☐ £ Report fly-tipping to the council quickly; most NI councils take reports online
- ☐ £ Organise or support community clean-ups, such as Live Here Love Here
- ☐ ££ Fix lighting, fencing and sightlines at dumping hotspots
- ☐ ££ Use a registered waste carrier and keep transfer notes: your duty of care doesn't end at the gate

NI: Councils enforce litter and fly-tipping law, and NIEA deals with larger illegal waste sites.
Take care: Don't move dumped waste that may be hazardous, such as asbestos or drums. Report it instead.

Misconnected drains → Right pipe, right place (Water)

Washing machines, sinks or yard drains plumbed into surface water drains send dirty water straight to rivers.

- ☐ £ Check where every drain on your site goes: surface water or foul sewer
- ☐ £ Dye-test suspect connections
- ☐ ££ Reconnect misconnected appliances to the foul sewer
- ☐ ££ Fit oil interceptors on yard drains where vehicles park or refuel

NI: Discharging trade effluent to the foul sewer needs NI Water consent. Polluting a waterway is an offence.

Light pollution → Light only what you need (Rewilding)

Unnecessary night lighting disrupts bats, insects and birds, and wastes energy.

- ☐ £ Switch off or dim lights when they aren't needed; use timers and motion sensors
- ☐ £ Point lights down and shield them; avoid lighting hedges, trees and water
- ☐ ££ Use warm-white LEDs (3000 K or lower)

NI: New lighting can need planning permission, and excessive artificial light can be a statutory nuisance.

Take care: Keep safety lighting where people need it.

Low soil carbon → Build carbon back in (Soil)

Soils low in carbon hold less water and fewer nutrients, and lose their structure.

- ☐ £ Keep the soil covered and growing for as much of the year as possible
- ☐ ££ Add compost and well-made digestate
- ☐ ££ Trial biochar from clean wood waste, mixed with compost
- ☐ ££ Turn some arable ground into grassland or herbal leys

NI: Spreading biochar on land may need a waste exemption or permit, depending on what it's made from.

Take care: Biochar quality varies. Buy certified material and don't expect results in one season.

Important. For contaminated land, forever chemicals (PFAS) or any spill reaching a drain, watercourse or groundwater, get specialist advice and involve the regulator. In NI, report pollution to the NIEA Incident Hotline on **0800 80 70 60** (24 hours). Contaminated-land remediation normally goes through planning, using the Land Contamination: Risk Management (LCRM) framework.

Sources. DAERA and CAFRE guidance on the Nutrients Action Programme and the hedge-cutting closed period (1 March to 31 August); NIEA incident reporting (nidirect); LCRM (Environment Agency, used by NIEA); UK and EU work on PFAS (Defra PFAS Plan 2026, ECHA); the All-Ireland Pollinator Plan; the "Don't Mow, Let It Grow" initiative run by NI councils and partners.

Position at 16 October 2026. For information, not legal or engineering advice. Robert J. Park · linkedin.com/in/robertparkuk · WhatsApp @robertjpark